

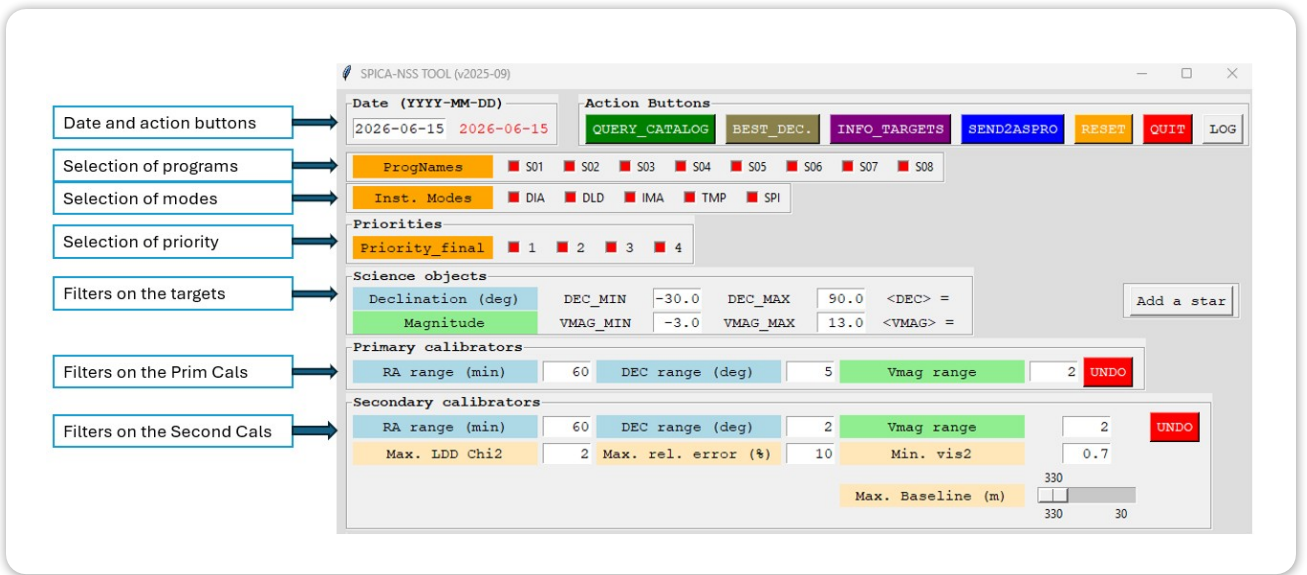
Description of the aims and functionalities of the Night Scheduling Software (NSS)

Denis Mourard, David Salabert, June 2026

Within the framework of the ERC Adv Grant (2021) called “Interferometric Survey of Stellar Parameters” (ISSP), our team has developed python tools aiming at managing over five years a large interferometric survey of angular diameters measurements with the CHARA Array, Mt Wilson USA, and combining the SPICA (R band), MIRC-X (H band), and MYSTIC (K band) instruments.

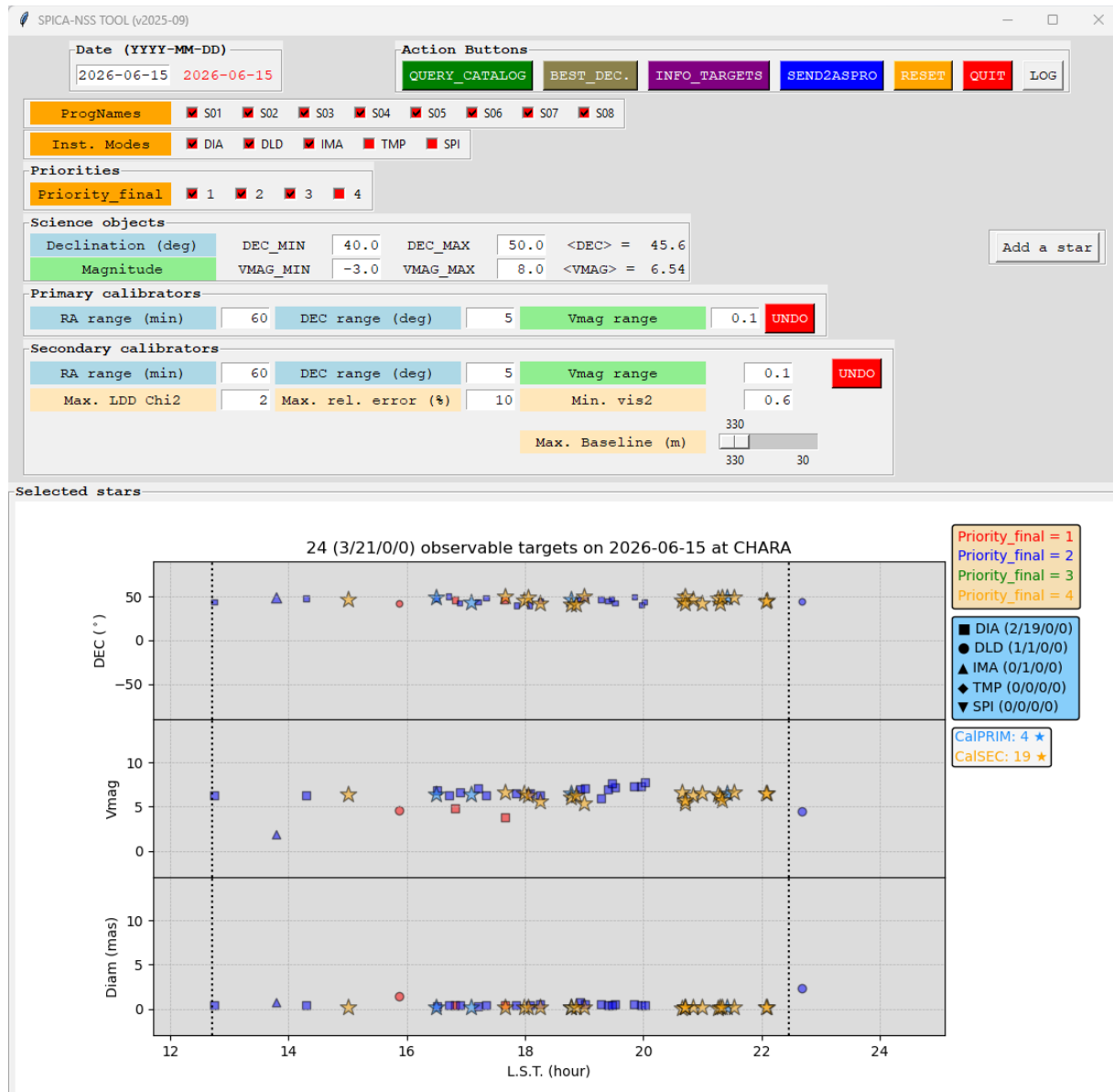
The **NSS** is based on a catalog called ‘spicadb’ containing about 3000 suitable targets for the different science purposes of the ISSP survey. Each entry in the catalog contains a spicadb ID, a valid simbad identifier, the piname, program name, comments on the target, a simple toy model compatible with ASPRO2, the CHARA and SPICA configuration, the priority, the completion rate, the HD number, coordinates, the already validated observing blocks, and different magnitudes.

The aim of the **NSS** is to select suitable observable targets for a certain date, considering completion rate, priority, and selection criteria like declination and magnitude range. It permits also to select some primary calibrators, previously identified with the JSDC catalog, as well as secondary calibrators with a direct search in the JSDC catalog, also including some additional filtering based on spectral types and on a control with the BadCal catalog.



*Entrance panel of the **NSS** software*

For a given observing night, the survey mode is organized around a small range of declination to avoid changes of configuration and to optimize the stability of the transfer function of the interferometric instruments. The selection of stars is done depending on their observability, their priority (including the level of the completion rate), and some additional filtering on the programs or on the observing modes. In the example below, 24 targets have been selected, 4 primary calibrators and 19 secondary calibrators. The button “Add a star” permits to add one spicadb targets that has been rejected by the filters if the PI requests this. The SEND2ASPRO permits to transfer to ASPRO2 all the required information, including the SPICADB identifier, as well as the PI_name, Prog_Name, and the simple toy model. These metadata are then transmitted through the A2P2 principles from ASPRO2 to the SPICA control software for correct referencing in the headers of the data files. At the end of the processing, the validation of the data files permits us to update the completion rate, for further considerations in the night scheduling. The export to ASPRO2 is also presented.



An example of night preparation

